

THE M-CATS

FALL 2026

Genetics and Evolution

BIOLOGY CHAPTER 12

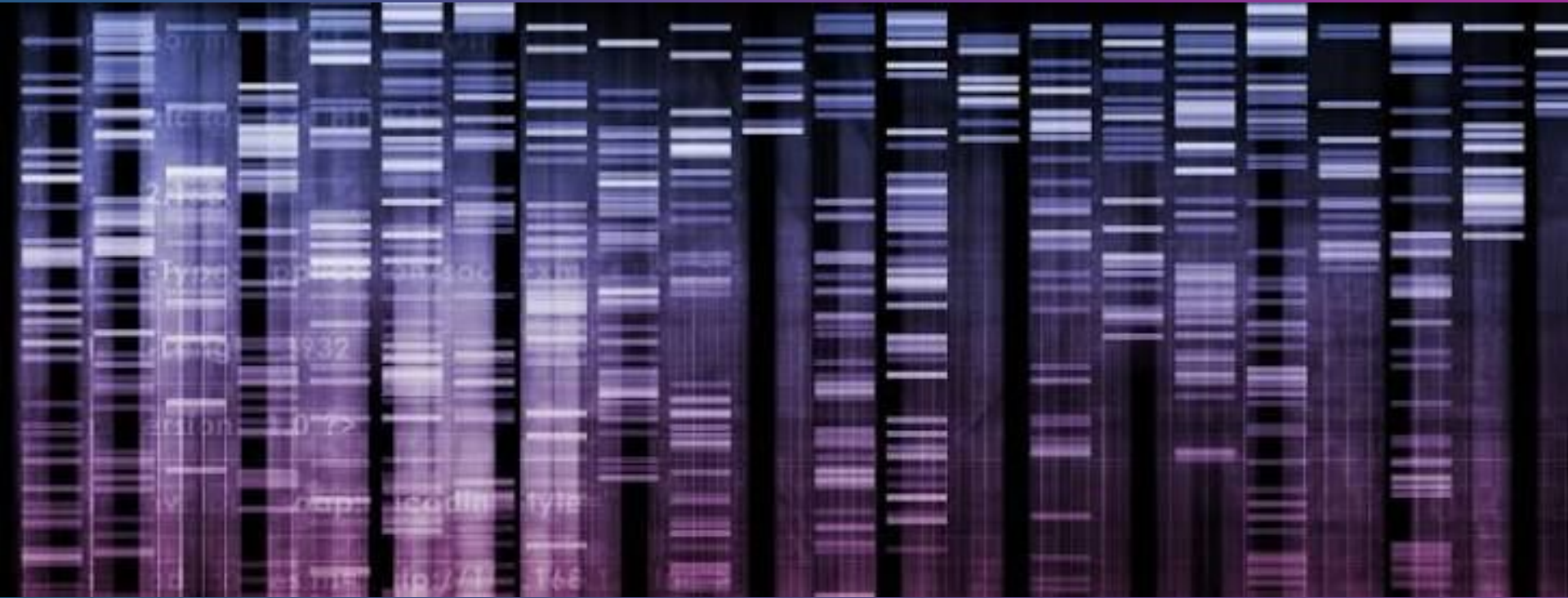
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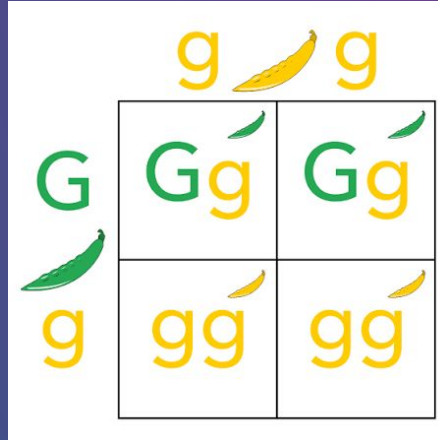
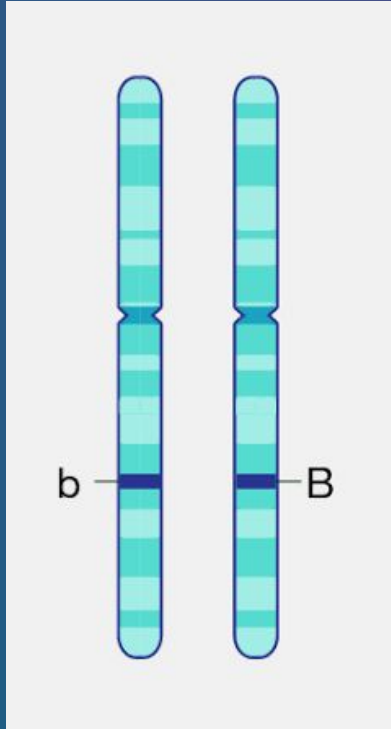
12.1

Fundamental Concepts of Genetics

Genes, Alleles, Geneotypes, and Phenotypes



Genes, Alleles, Genotypes, and Phenotypes



Words to Know: Locus, Homozygous, Heterozygous, Hemizygous

Mendelian Genetics



Core Tenets of Mendelian Genetics

- Segregation
 - Alleles inherited from parents segregate during meiosis; gametes only carry one allele
 - If inherited alleles are different, only one will be expressed....*not really*
- Independent assortment
 - Inheritance of one gene does not affect inheritance of another... *again.. not always*

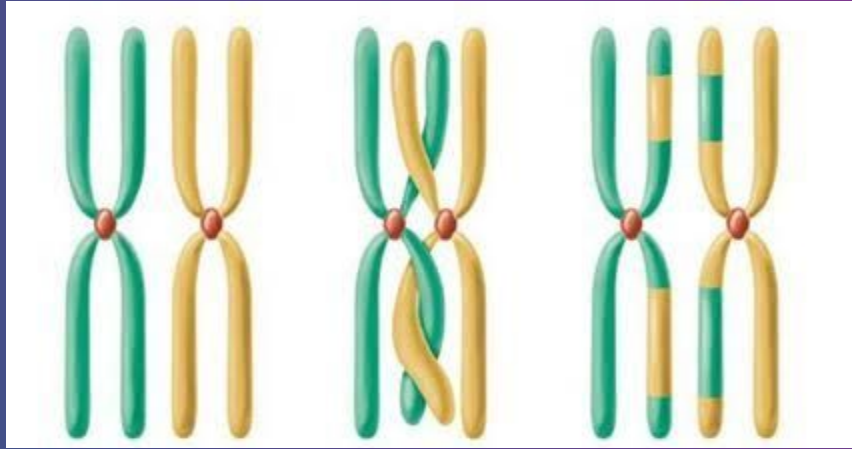
NonMendelian Genetics



- Codominance
 - Blood types
- Incomplete Dominance
 - Snapdragons



Independent (and non independent) assortment and DNA



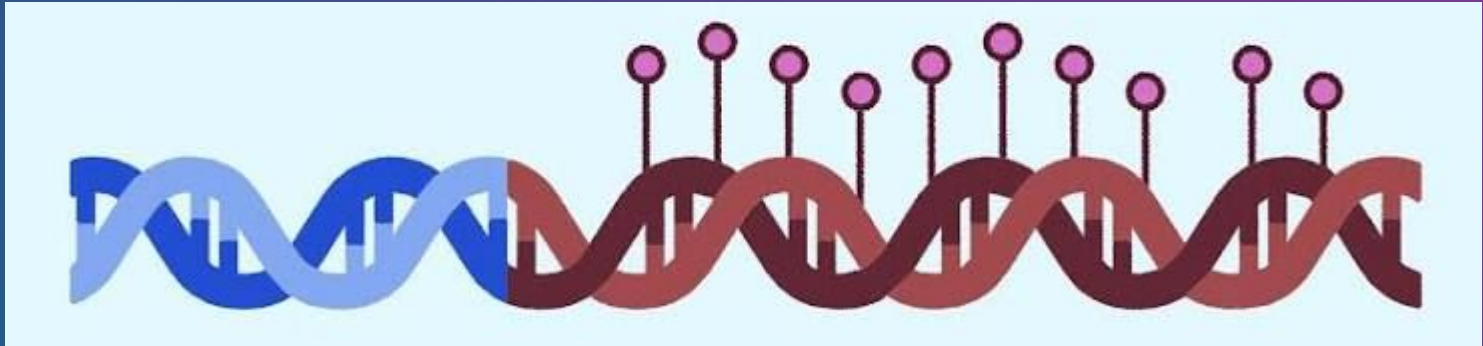
Chromatids, Chromosomes, Tetrads, and Recombination

Genes aren't always inherited independently, ...*but more on that later*

Fundamental unit of inheritance

DNA is the unit of inheritance*

Epigenetics, Methylation, Barr Bodies, and Imprinting



Quiz

A researcher breeds two homozygous plants, one blue (f) and one red(F). Which of the following matches the phenotype with the likely inheritance pattern?

- a. Blue, codominance
- b. Purple, Mendelian dominance
- c. Red, incomplete dominance
- d. Purple, Incomplete dominance

An autosomal recessive disease has an allele frequency of 3×10^{-3} . What is the likelihood that an individual in the population has the disease?

- a. 9×10^{-5}
- b. 9×10^{-6}
- c. 3×10^{-6}
- d. 3×10^{-3}

12.2

Changes in the Gene Pool

2 Quick Terms

- **Penetrance** (Population)
 - How likely a genotype will express a certain phenotype
- **Expressivity** (Individual)
 - Constant or varied phenotypes

Mutations

- Nucleotide level mutations
 - Point mutations
 - Silent
 - Missense
 - Nonsense
 - Frameshift mutations
- Chromosomal mutations
 - Deletion
 - Duplications
 - Inversion
 - Insertion
 - Translocation

cDNA Codon Table

Second Position

		Second Position				
		T	C	A	G	
First Position	T	TTT } Phe TTC } TTA } Leu TTG }	TCT } Ser TCC } TCA } TCG }	TAT } Tyr TAC } TAA } STOP TAG } STOP	TGT } Cys TGC } TGA } STOP TGG } Trp	T C A G
	C	CTT } Leu CTC } CTA } CTG }	CCT } Pro CCC } CCA } CCG }	CAT } His CAC } CAA } Gln CAG }	CGT } Arg CGC } CGA } CGG }	T C A G
	A	ATT } Ile ATC } ATA } Met ATG }	ACT } Thr ACC } ACA } ACG }	AAT } Asn AAC } AAA } Lys AAG }	AGT } Ser AGC } AGA } Arg AGG }	T C A G
	G	GTT } Val GTC } GTA } GTG }	GCT } Ala GCC } GCA } GCG }	GAT } Asp GAC } GAA } Glu GAG }	GGT } Gly GGC } GGA } GGG }	T C A G



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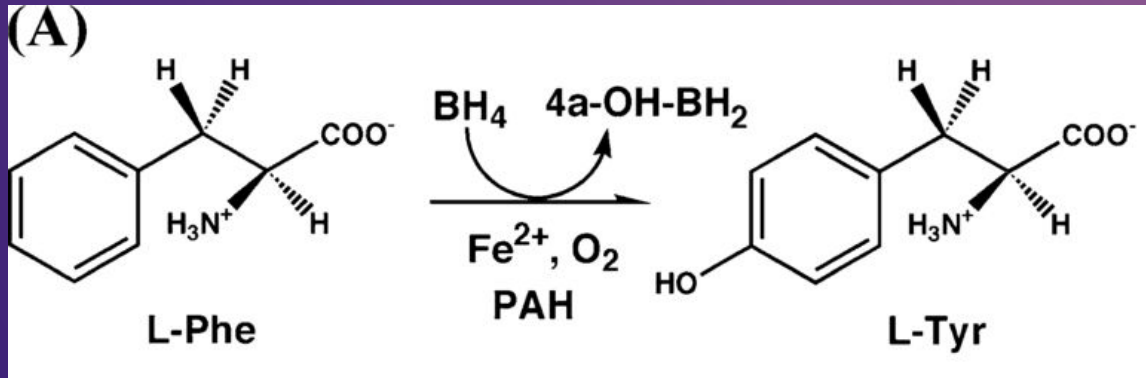
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Types of Mutations

- Advantageous
 - Industrial Melanism
- Deleterious
 - Inborn errors of metabolism
 - PKU-Phenylalanine Hydroxylase
 - *Don't memorize this lol*



Genetic exchange in population

- Genetic Leakage
 - Species 1 + Species 2
 - Hybrid + Species 1
 - The offspring is species 1 offspring, but some species 2 DNA
- Genetic Drift
 - Changes in composition of gene pool due to change
 - Founder Effect
 - Bottlenecks

12.3

Analytical Approaches In Genetics

Different Techniques to Know and Concepts to Cover

- Punnett squares
 - P, F₁, F₂ generations
 - Hybrid, dihybrid, Sex-linked crosses
 - Test Crosses
- Hardy-Weinberg equilibrium
 - No Mutations, Random Mating, No Gene flow, Large Population, No Advantage from genotype
 - $p+q=1$
 - $p^2+2pq+q^2=1$
- Linked Genes
 - Gene map
 - Centimorgans

12.3

Questions

Questions

In a population at Hardy-Weinberg equilibrium, an autosomal recessive condition affects 1 in 2,500 individuals. What fraction of the population are unaffected carriers?

- A. 1/50
- B. 1/2500
- B. 49/2500
- C. 98/2500

A woman who is a carrier for an X-linked recessive disorder has children with an unaffected man. What proportion of their daughters are expected to be affected, and what proportion of their sons?

- A. 1/2 of daughters, 1/2 of sons
- B. 0 of daughters, 1/2 of sons
- C. 1/4 of daughters, 1/4 of sons
- D. 0 of daughters, all sons

Two genes are located on the same chromosome. In a testcross, recombinant offspring occur at a frequency of 8%. What is the map distance between the genes, and what does this imply?

- A. 8 map units; the genes are relatively close together
- B. 8 map units; the genes assort independently
- C. 92 map units; the genes are far apart
- D. 16 map units; the genes are close together

Questions

A point mutation changes a codon from GAG (glutamate) to GTG (valine). Which description of this mutation's molecular consequence is most accurate? (Also, do you think this would likely have an effect on the protein's tertiary structure?)

- A. A silent mutation
- B. A missense mutation
- C. A nonsense mutation
- D. A frameshift mutation

A person with genotype $I^A I^B$ has AB blood type, expressing both A and B antigens on their erythrocytes. A snapdragon cross between a red-flowered and a white-flowered plant yields pink offspring. These two cases illustrate, respectively:

- A. Incomplete dominance and codominance
- B. Codominance and incomplete dominance
- C. Complete dominance and codominance
- D. Codominance and complete dominance

Which of the following describes incomplete penetrance rather than variable expressivity?

- A. Individuals carrying a disease allele show symptoms ranging from mild to severe
- B. Some individuals carrying a disease-causing genotype show no disease phenotype at all
- C. Heterozygotes display an intermediate phenotype between the two homozygotes
- D. Both alleles are fully and simultaneously expressed in the heterozygote

Questions

An X-linked recessive disorder has an allele frequency of $q = 0.08$ in a population at Hardy-Weinberg equilibrium. What are the expected frequencies of affected males and affected females, respectively?

- A. 0.64% and 8%
- B. 8% and 0.64%
- C. 8% and 8%
- D. 16% and 0.64%

A geneticist performs crosses to map three linked genes and obtains the following recombination frequencies: A-B = 16%, B-C = 31%, A-C = 15%. What is the correct gene order?

- A. C - A - B
- B. A - C - B
- C. A - B - C
- D. The genes are unlinked and cannot be mapped

Two mutations occur in the coding region of a gene. Mutation 1 deletes three consecutive nucleotides; Mutation 2 deletes two consecutive nucleotides. Which outcome is expected?

- A. Both mutations shift the reading frame, but the two mutations together restore the reading frame
- B. Mutation 1 removes one amino acid while preserving the downstream reading frame; Mutation 2 shifts the reading frame
- C. Mutation 1 removes one amino acid; Mutation 2 is silent because it removes fewer nucleotides
- D. Neither mutation affects the protein, since deletions only affect noncoding regions

12.3

Evolution

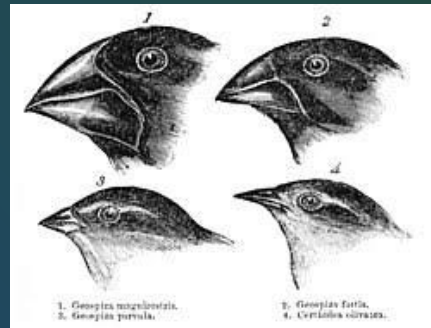
Theories of evolution

- **Natural selection**
 - Organisms with advantageous (favorable) mutations for making it to reproduction will pass these genes on to offspring (High **Fitness**)
 - Black Widows
- **Neo Darwinism**
 - Favorable traits become more common in gene pools
 - Inclusive fitness- Number of offspring and Support of those offspring
- **Punctuated Equilibrium**



Types of natural selection and directional evolution

- **Stabilizing**
 - Selects for the average phenotype
- **Directional**
 - Selects for one extreme
- **Disruptive**
 - Selects for both extremes, selects against the average



Speciation

- Development of new species through evolution
- Isolation can cause speciation
 - Prezygotic mechanisms
 - Allopatric
 - Temporal
 - Ecological
 - Sympatric
 - Behavioral
 - Reproductive
 - Gametic
 - Postzygotic
 - Hybrid inviability or sterility



Patterns of evolution

- Convergent
 - Unrelated or distantly related species evolving similar traits in response to similar environmental pressures
- Divergent
 - Closely related species evolve away from each other due to different environmental pressures
- Parallel
 - Related species evolve similarly in response to similar environments



Evolutionary time can be measured by DNA similarity (**Molecular Clock Model**)

Questions

A small group of individuals becomes isolated from a large mainland population after a storm. Over subsequent generations, allele frequencies in the isolated group diverge markedly from the mainland, including fixation of some alleles that were rare on the mainland. Which process best explains this?

- A. Natural selection favoring the previously rare alleles
- B. Genetic drift following a founder event
- C. Gene flow between the two populations
- D. Increased mutation rate in the isolated group

In a bird species, individuals with intermediate beak sizes are best able to exploit the seeds available on an island, while both very small and very large beaks are disadvantageous. Over generations, what change in the distribution of beak size is expected? (What kind of selection is this?)

- A. The mean shifts toward larger beaks, with variance unchanged
- B. The mean remains approximately constant while variance decreases, as extreme phenotypes are selected against
- C. The distribution becomes bimodal, with two distinct peaks
- D. Both the mean and the variance increase

Which individual has the highest evolutionary fitness, as the term is used in population biology?

- A. A male who is the largest and strongest in his population but leaves no surviving offspring
- B. A male of average size who leaves six offspring that survive to reproduce
- C. A female who lives longer than any other individual in the population but produces two offspring, one of which survives to reproduce
- D. A male with the greatest resistance to local parasites who dies before reaching reproductive age

Questions

A river permanently changes course, dividing a single population of rodents into two groups that cannot cross. After many generations the two groups are brought back into contact and no longer interbreed successfully. This sequence illustrates:

- A. Sympatric speciation
- B. Allopatric speciation
- C. Gene flow maintaining a single species
- D. Polymorphism within a single population

A conservation program maintains an endangered species in a small captive colony where close relatives frequently mate. Over several generations, managers observe increased incidence of rare recessive disorders. What is happening at the population-genetic level?

- A. Inbreeding is changing allele frequencies by introducing new deleterious mutations
- B. Inbreeding is increasing the proportion of homozygotes relative to Hardy-Weinberg expectations, so pre-existing rare recessive alleles are more often paired and expressed, without necessarily changing allele frequencies themselves
- C. Inbreeding increases heterozygosity, masking recessive alleles
- D. Outbreeding depression is occurring

Researchers estimate the divergence time of two lineages using two independent approaches: counting the number of neutral nucleotide substitutions that have accumulated between their genomes, and radiometric dating of fossils using an isotope with a known half-life. Which statement best characterizes these two methods?

- A. Both measure the same physical process, so they cannot serve as independent checks on each other
- B. Both rely on processes that are separate and random, but provide approximations that are likely accurate
- C. Radiometric dating depends on the chemical reactivity of the isotope, while molecular dating depends on natural selection favoring specific substitutions
- D. Neutral substitutions accumulate faster in lineages under stronger natural selection